

Review on geometric optics and interference

Phys2500J

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Overview

1. Reflection and Refraction

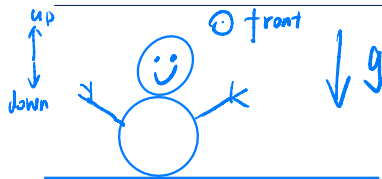
Many Conceptual Problem.

2. Dispersion

3. Polarization

4. Interference

Question?

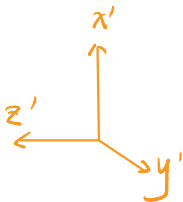
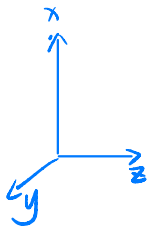


$$\underline{\text{left}} = \vec{up} \times \vec{front}$$

pseudo vector

$$\Rightarrow \vec{left}' = -(\vec{up}' \times \vec{front}')$$

Why the image in the mirror is opposite in the left and right instead of up and down?



symmetry operation

$$\begin{cases} x' = x \\ y' = y \\ z' = -z \end{cases}$$

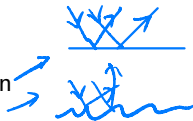
Reflection and Refraction

- Reflection

- Specular reflection

- Diffuse reflection

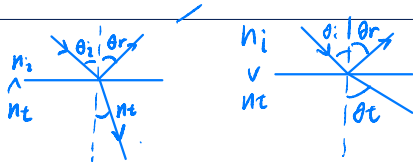
- Laws of reflection $\theta_i = \theta_r$



- Refraction

- Snell's Law: $n_i \sin(\theta_i) = n_t \sin(\theta_t)$

- Relation between index and the wavelength



299792458 m/s.

$$v = \frac{c}{n}$$

$$v = \lambda f$$

f be same. $\Rightarrow \lambda' = \frac{\lambda_0}{n}$

- Huygens's principle: check the definition on slide.

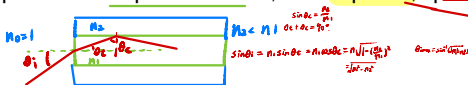
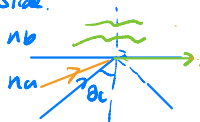
- Total internal reflection

- Only happens if $n_a > n_b$

- Critical Angle $\theta_{critical} = \arcsin(n_b/n_a)$

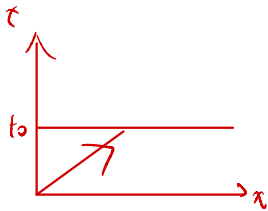
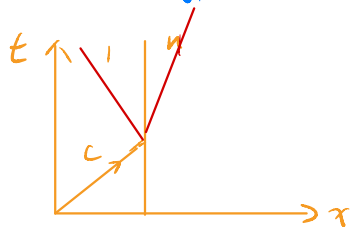
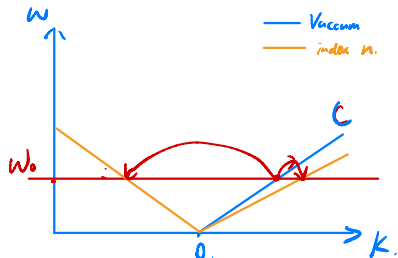
- Application: Optical Fibers, Pentaprism, Sparkle of diamonds

$$n_a \sin(\theta_a) = n_b \sin(\theta_b)$$



More thoughts on refraction

Band Structure.



$$c = 299792458 \text{ m/s.}$$

$$V_{\text{phase}} = \frac{\omega}{k} = \underline{V_{\text{group}} = \frac{d\omega}{dk}}$$

$$\underline{V_{\text{phase}} = \frac{c}{n}.}$$

Dispersion

$$\sin \theta_i = n \sin \theta_t$$

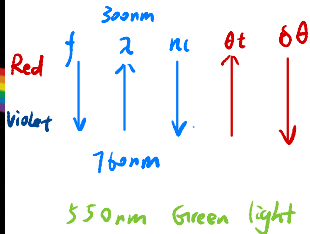
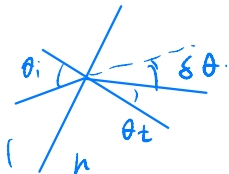


Figure: Dark side of the moon–Pink Floyd

Dispersion

Definition

The index of refraction depends on the wavelength of the light

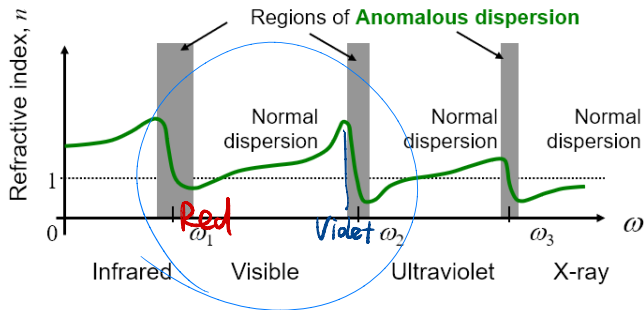


Figure: Refractive Index vs. Frequency

Examples for dispersion

- ① times of reflection
- ② sequence of colors
- ③ Intensity

Examples

Rainbows

Point out the different between Primary rainbow and Secondary rainbow?

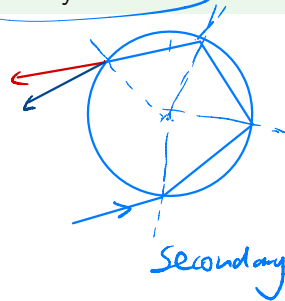
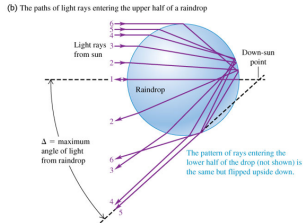
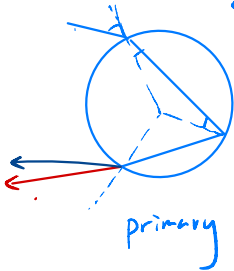
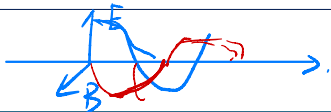


Figure: Path of light rays entering the upper half of a raindrop

Polarization



Definition

An electromagnetic wave is linearly polarized if the electric field has only one component.

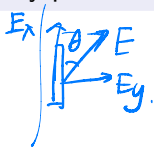
- Polarizing filter, polarizer

- Malus's Law $I(\theta) = I_0 \cos^2 \theta$

- Brewster's Angle

- Application: 3D glasses, sunglasses

- Birefringence: halfwave plate, quarter-wave plate

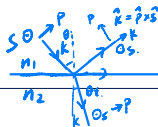
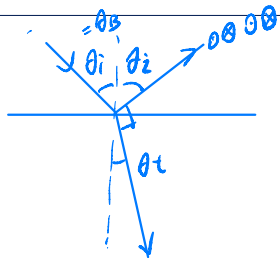


$$I \propto \langle E^2 \rangle$$



$$\Delta L = (n_o - n_e)d$$

Brewster's Angle



$$\begin{cases} D_{in} = D_{2n}, & E_{it} = E_{rt} \\ B_{in} = B_{2n}, & H_{it} = H_{rt} \end{cases}$$

$$r_p = \frac{n_2 \cos(\theta_i) - n_1 \cos(\theta_t)}{n_2 \cos(\theta_i) + n_1 \cos(\theta_t)} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \xrightarrow{r_s, t_s, t_p} 0.$$

$$\theta_i + \theta_t = 90^\circ, \quad r_p \rightarrow 0.$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t = n_2 \cos \theta_i \Rightarrow \tan \theta_i = \frac{n_2}{n_1}$$

$$\Rightarrow \theta_i = \arctan\left(\frac{n_2}{n_1}\right) = \theta_B.$$

reflect wave is linearly polarized.

Scattering

Why clouds are white? (seems like questions from One Hundred Thousand Whys)

Reasons: Clouds are white because they scatter all wavelengths efficiently

Overview

Key concepts of interference

$\vec{E} = A(r) \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)}$

$I \propto \hat{E} \cdot \hat{E}^*$

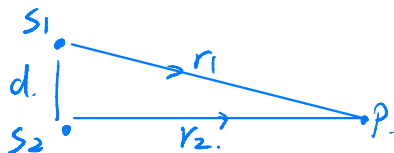
- Wave function. (Recommend written in Euler function form)
- Superposition principle $\sum E_i$
- Phase difference & Optical Pathlength difference (OPD)

$$\delta = \Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta L$$

$\downarrow$
 λ in vacuum

$\Delta L = n \cdot \text{distance}$

Two-source interference



plane wave. $A(\nu) = \text{const.}$

$$E_p = E_1(r_1) + E_2(r_2)$$

$$= E_1 \cdot e^{i(kr_1 - \omega t)} + E_2 \cdot e^{i(kr_2 - \omega t)}$$

$$= e^{-i\omega t} (E_1 e^{ikr_1} + E_2 e^{ikr_2})$$

$$I \propto E_p \cdot E_p^* = (e^{-i\omega t} (E_1 e^{ikr_1} + E_2 e^{ikr_2})) \cdot (e^{i\omega t} (E_1 e^{-ikr_1} + E_2 e^{-ikr_2}))$$

$$= | (E_1^2 + E_2^2 + E_1 E_2 e^{ik(r_1 - r_2)} + E_1 E_2 e^{-ik(r_1 - r_2)}) |$$

$$= E_1^2 + E_2^2 + 2E_1 E_2 \cos(k(r_1 - r_2))$$

$$\delta = k(r_1 - r_2)$$

$$= I_1^2 + I_2^2 + 2\sqrt{I_1 I_2} \cos(\delta)$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Two-source interference intensity

Result

- Constructive interference: $\delta = 2m\pi$
- Destructive interference: $\delta = (2m + 1)\pi$

$m \in \mathbb{Z}$.

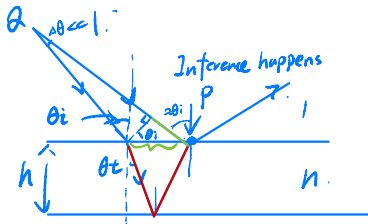
Intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\delta)$$

Fringe visibility

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} \leq 1$$

Thin film interference



$$\theta_i = 0 \quad \Delta L = 2nh$$

$$OPD = R - G \quad \therefore = \frac{2nh}{\cos \theta_t} (1 - \sin^2 \theta_t) = 2nh \cos \theta_t \left(1 + \frac{\pi}{2}\right)$$

$$R: n \cdot \left(\frac{2h}{\cos \theta_t}\right)$$

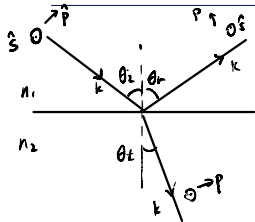
$$G: (2h \cdot \tan \theta_t) \cdot \sin \theta_i = 2h \frac{\sin^2 \theta_t}{\cos \theta_t} \cdot n$$

$$n \sin \theta_t = \sin \theta_i \quad \text{Snell's law}$$

$$n_1 < n_2$$

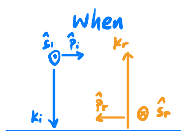
$$\leftarrow \begin{array}{c} \text{---} \\ \text{---} \end{array} \left| \begin{array}{c} +\pi \\ \text{---} \end{array} \right. \begin{array}{c} n_1 \\ n_2 \end{array}$$

π phase shift



$$\left\{ \begin{array}{l} r_p = \frac{n_2 \cos(\theta_i) - n_1 \cos(\theta_t)}{n_2 \cos(\theta_i) + n_1 \cos(\theta_t)} \\ r_s = \frac{n_1 \cos(\theta_i) - n_2 \cos(\theta_t)}{n_1 \cos(\theta_i) + n_2 \cos(\theta_t)} \end{array} \right.$$

consider $\theta_i = 0^\circ$, $\theta_t = 0^\circ$.



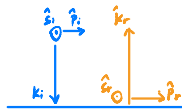
$$r_p = \frac{n_2 - n_1}{n_2 + n_1} > 0.$$

$$r_s = \frac{n_1 - n_2}{n_2 + n_1} < 0, \text{ opposite.}$$

When $n_1 > n_2$

$r_p < 0$. opposite

$r_s > 0$.



Interference model

$$D \gg \lambda \quad D \gg d.$$



$$\frac{a}{d} = \frac{\lambda}{D} \Rightarrow \Delta L = \frac{\lambda d}{D} = n\lambda.$$

$$n\text{th bright line locates?} \Rightarrow \pi = \frac{D}{d} n\lambda.$$

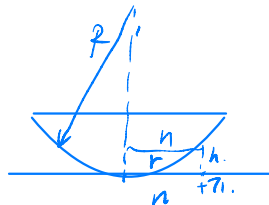
- Wavefront Splitting

- Young's Double-slit interferometer
- Fresnel's biprism
- Lloyd's mirror



- Amplitude Splitting

- Dielectric Film: Double-Beam Interference
- Fringe of Equal Inclination: Haidinger Fringes
- Fringe of Equal Thickness: Newton Rings
- Michelson Interferometer



$$\Delta L = 2h = 2(R - \sqrt{R^2 - r^2}) = \frac{r^2}{R} = n\lambda.$$

$$r \ll R$$

$$\Rightarrow r_n = \sqrt{n\lambda R}. \quad n\text{th dark line.}$$



$$\Delta L = 2 |X_2 - X_1|.$$

Problem Set

Onsite evaluation students' feedback



扫一扫